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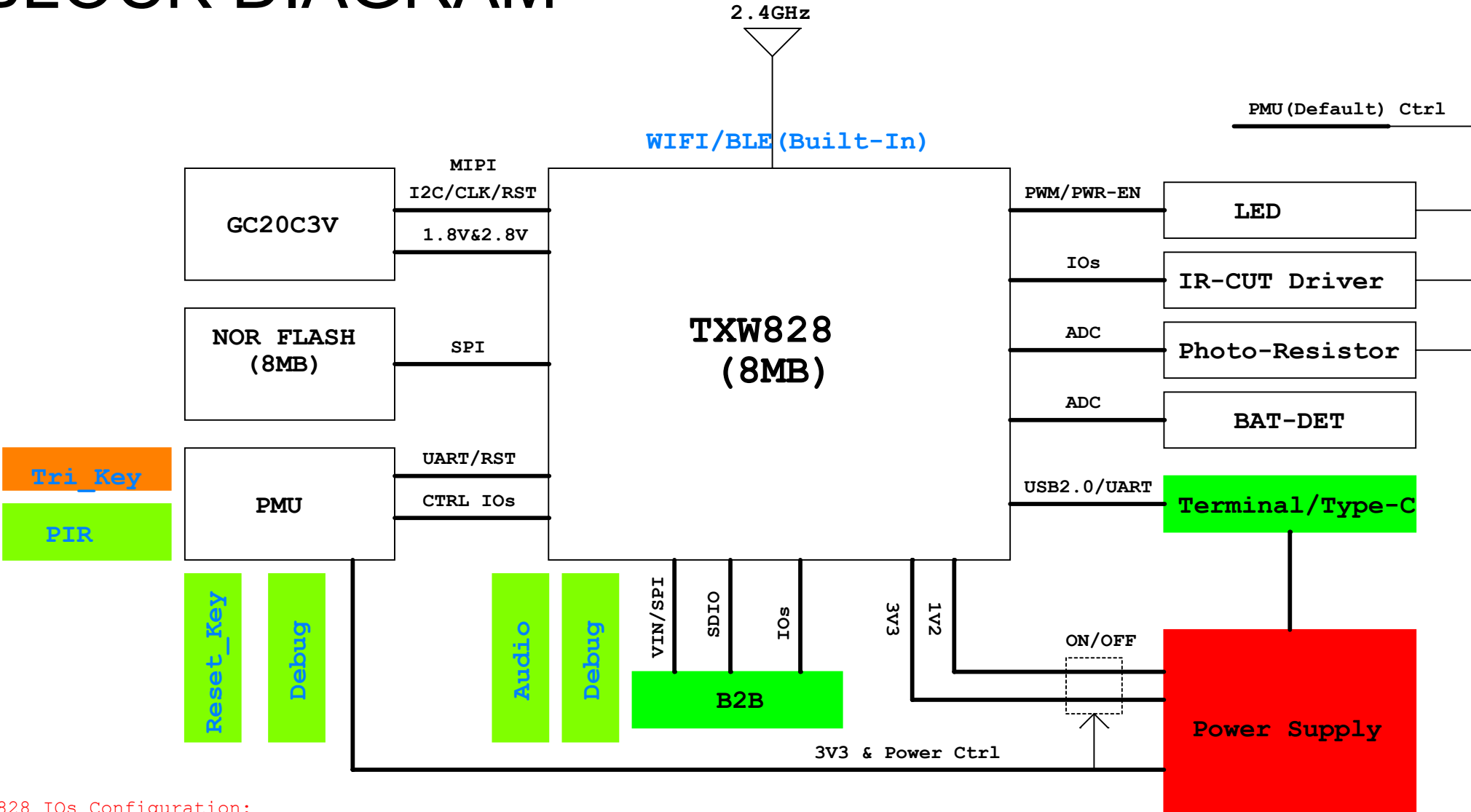
DESIGNED	Tino
REVISER	
REV	V1.0

- D
01. MAIN PAGE
02. CHANGE LIST
03. BLOCK DIAGRAM
04. POWER TREE
05. TXW82X
06. NS32L40X
07. POWER
- C08. GC20C3V
09. MISC
10. IR-CUT

Revision History

REV. CODE	DATE	Description
V1 0	2026/07/08	New Design

BLOCK DIAGRAM



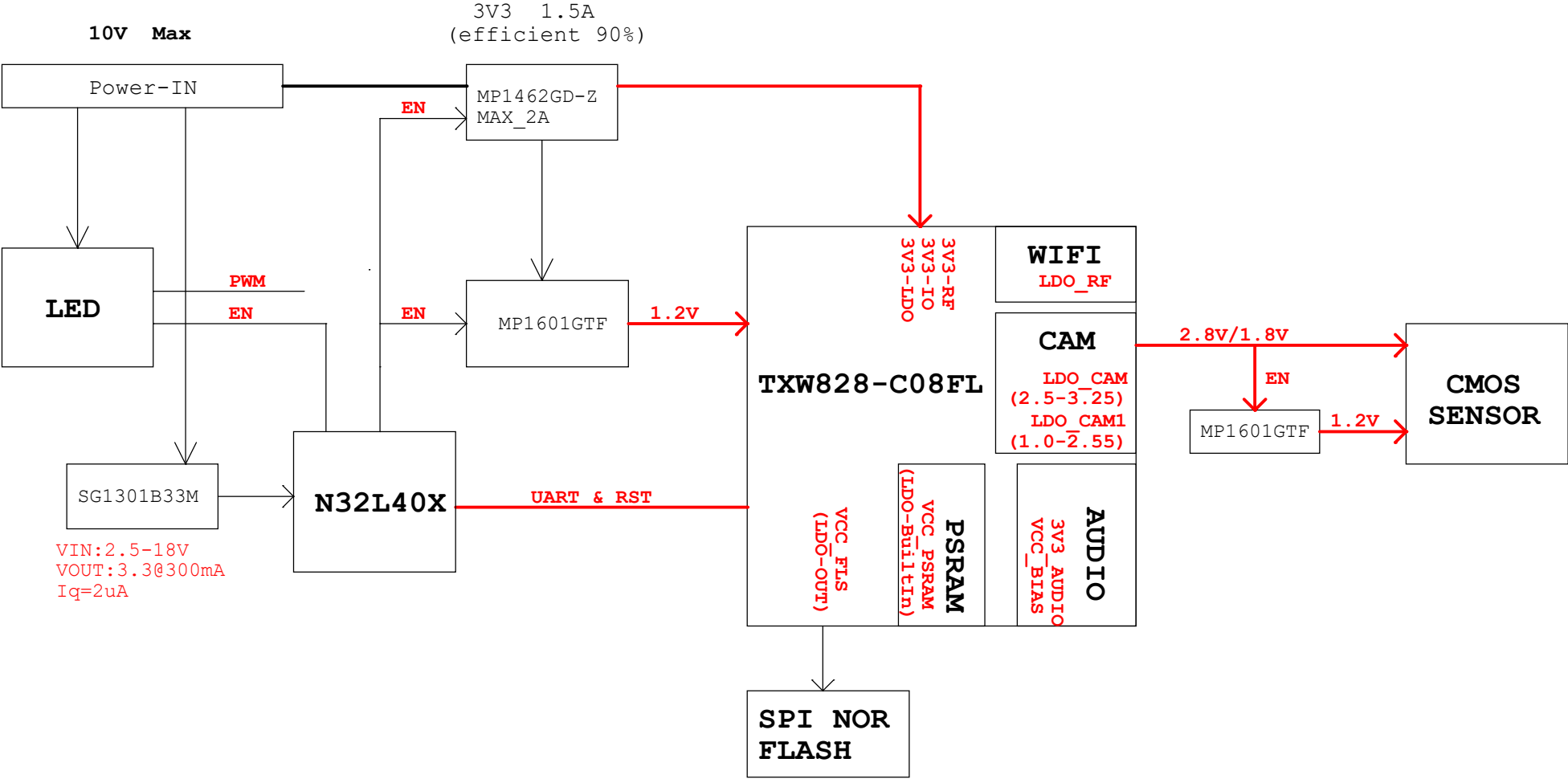
TXW828 IOs Configuration:

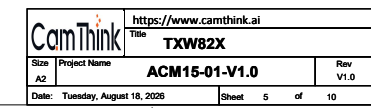
WiFi-Halow: SPI+Busy/Int/Wake/RST+Module_PWR
GC20C3V: MIPI*1+ I2C+RST+CLK+PDN+CAM_EN
SDIO: 1Bit+TF_EN+CD
Audio: Audio Input+Output
LED: PWM+CAM_EN+IR-CUT+PR_ADC
BATTERY: BAT_ADC
USB: *1 UVC
UART: *3 Debug, Cat-1, PMU-MCU
QSPI: 1.8V
OSC: 40M
OSC: 32.768K

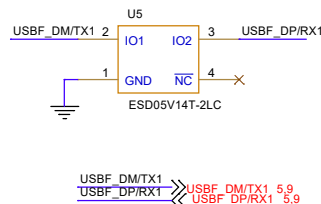
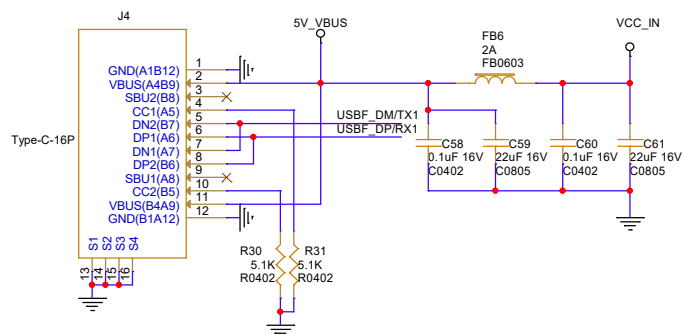
PMU Config:

- * UART to TXW82x
- * UART for Debug
- * Trigger Button
- * PWR RST
- * TXW82*_RST
- * Alarm(or PIR)
- * Module_PWR
- * SYS_PWR_EN
- * RTC

POWER TREE





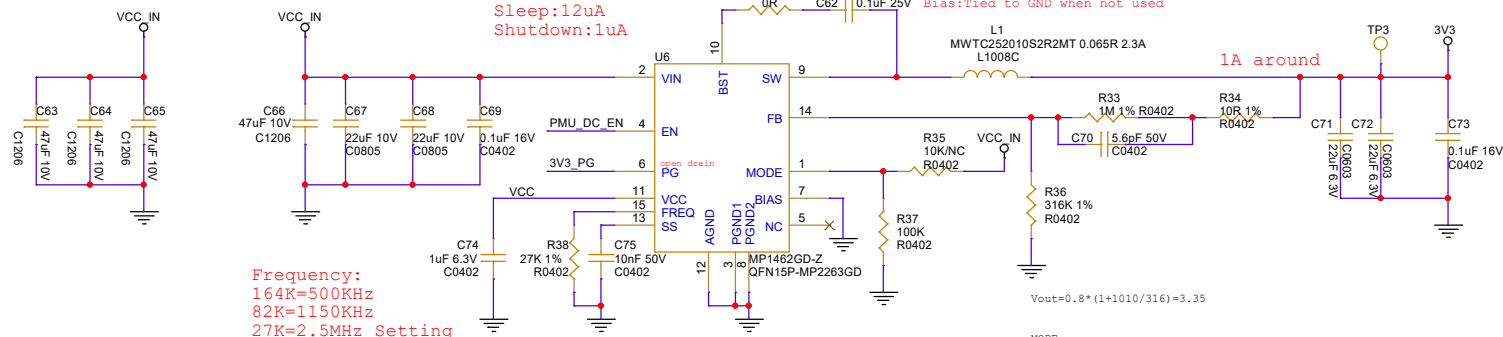


Vin Max=10V

VIN-->3V3

3.3-19V 2A
Sleep:12uA
Shutdown:1uA

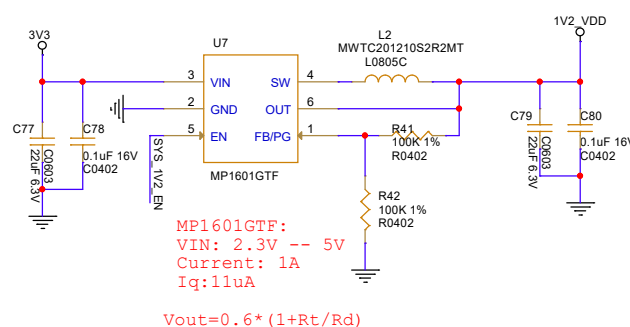
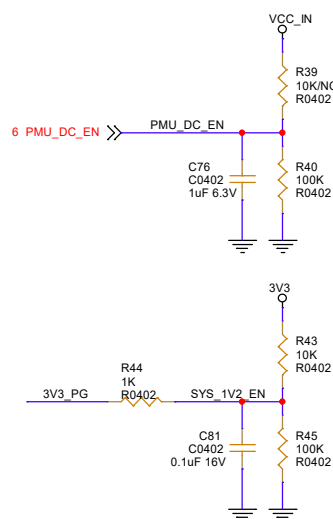
IC Powered:
LD01: By Vin, >5V, Full Regulation
LD02: By Bias, should up to 4.8V, then VCC internal circuit are powered by BIAS
(LD02 enabled when Bias>4.8V, then LD01 Disable)
when Vout=5V, Bias connect to 5V, for improved efficiency
Bias:Tied to GND when not used



Frequency:
164K=500KHz
82K=1150KHz
27K=2.5MHz Setting

Mode: (1.5M IPD)
H=PWM (Vcc-0.4)
L=PFM/PWM

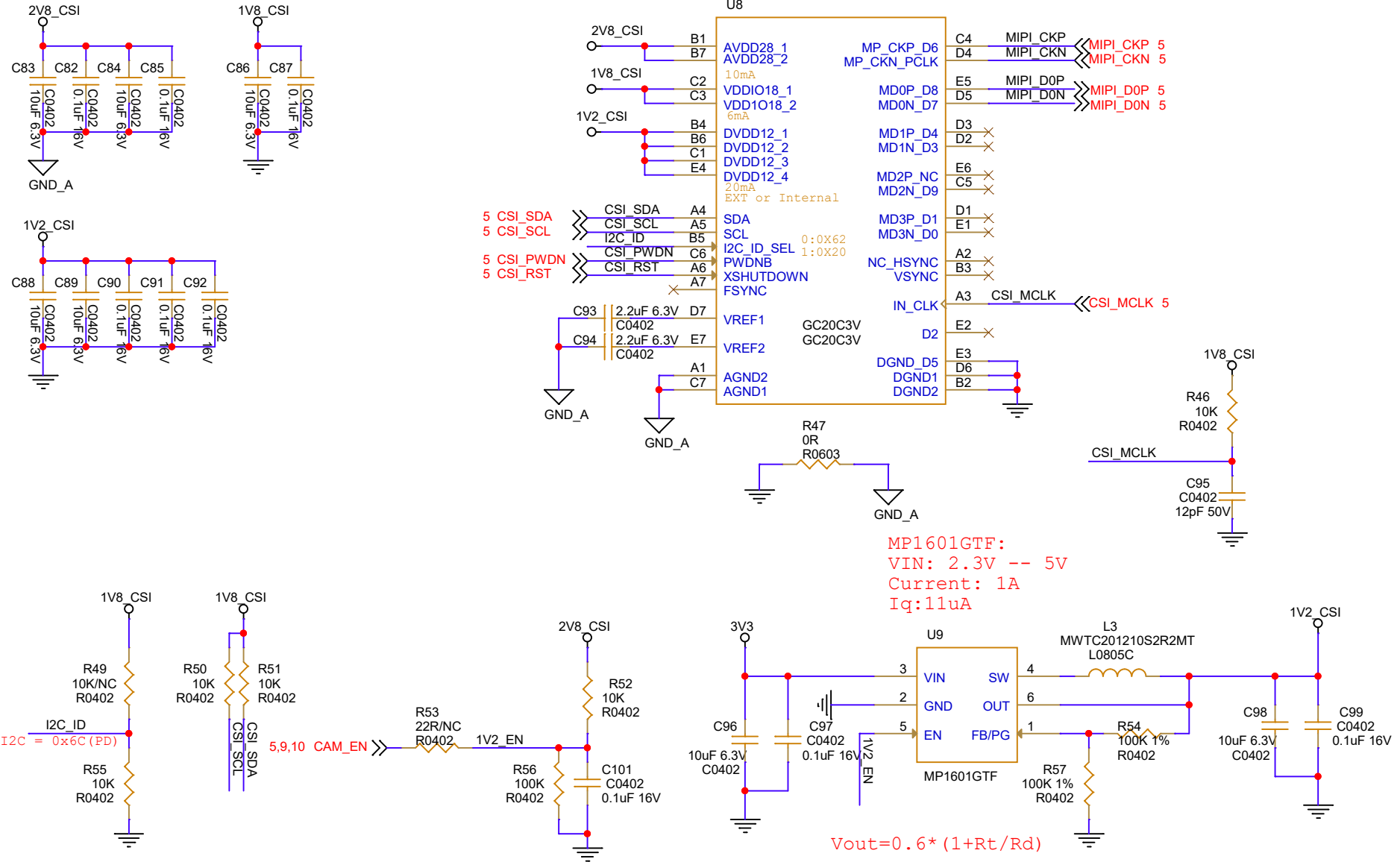
MODE:
LOW = PFM/PWM
HIGH = WM



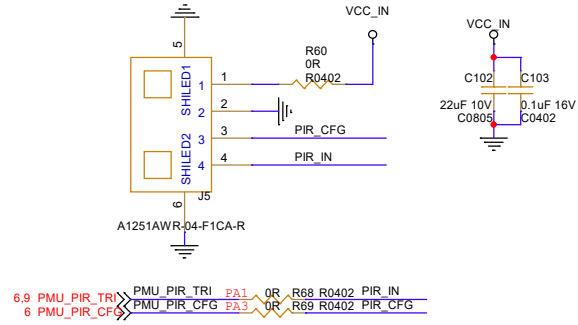
Vout=0.6*(1+Rt/Rd)

CamThink		https://www.camthink.ai	
Size		Title	
A3		Power	
Project Name		ACM15-01-V1.0	
Date		Tuesday, August 18, 2026	
Sheet		7 of 10	
Rev		V1.0	

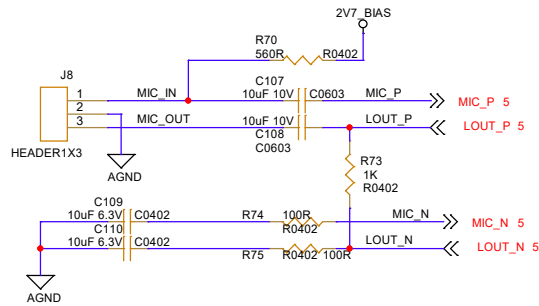
Sensor



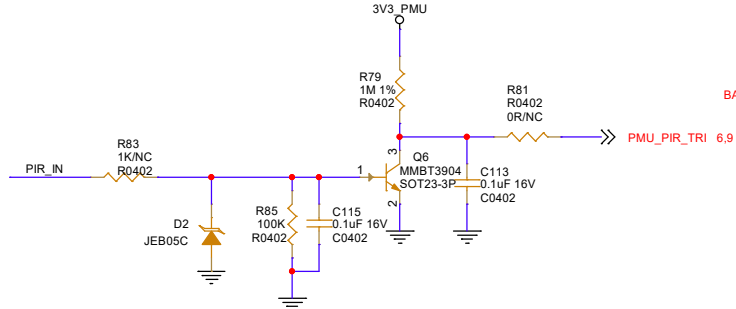
D



C

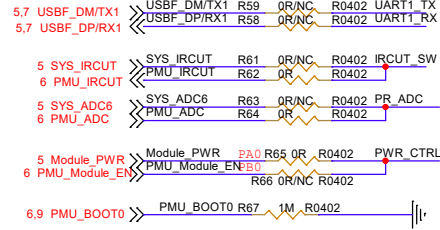


B

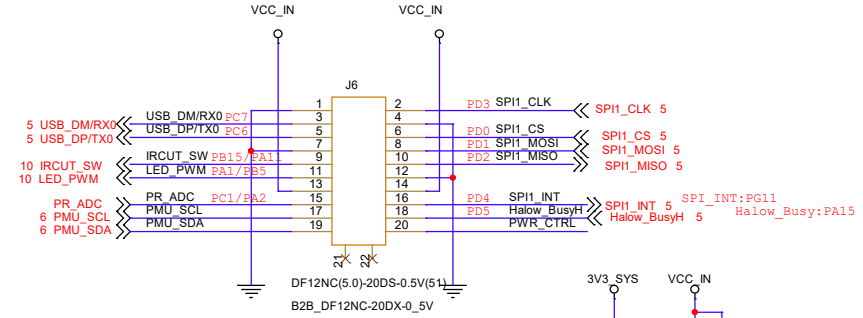


A

4

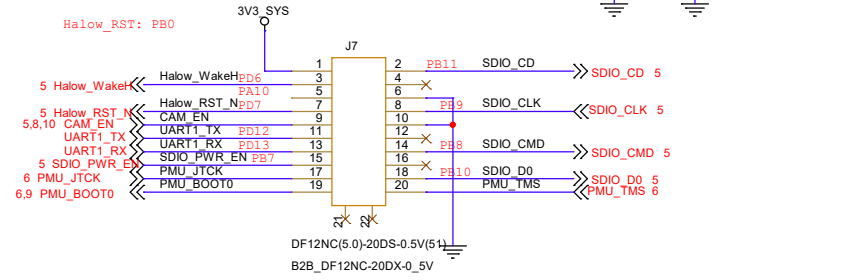


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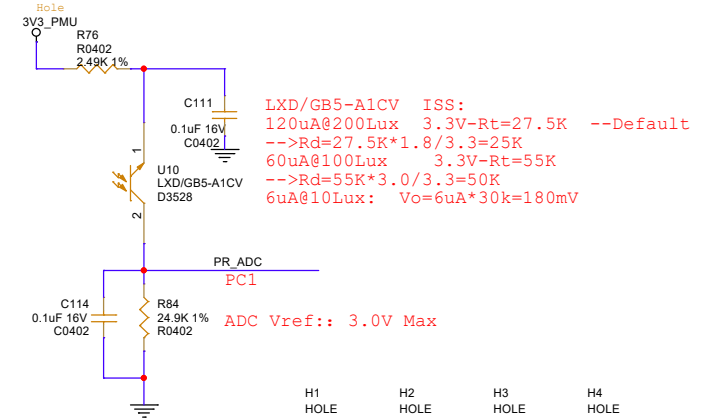
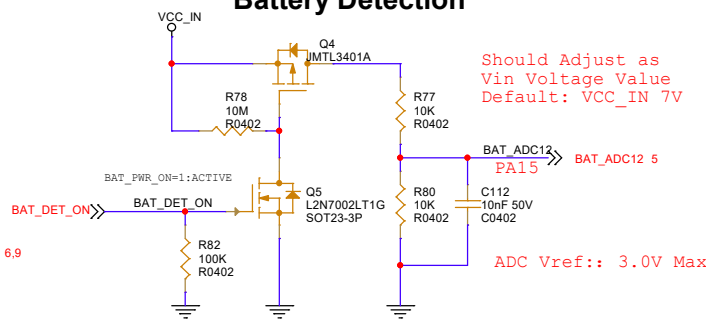
D

C



B

Battery Detection



A

D

C

B

A

